Hotel Room Reservation System – Project Report

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Abstract: This report documents the ongoing development of the Hotel Room Reservation System, focusing on architectural and functional enhancements made since the previous iteration, as well as new progress on physical and logical architectures. It establishes a clear structure for deployment and internal component interactions. A comprehensive set of UML diagrams is included to illustrate the system's static and dynamic aspects. The report presents elements to provide a complete view of the system's design progress.

Keywords: Domain Model, Functionalities, Class, Architecture

1. Introduction

This report presents the updated design and implementation progress of our Hotel Room Reservation System. Since the last iteration, significant modifications have been made, which have impacted on the progress of our work in a significant way. These changes have been suggested to us by our teachers and led to a deeper understanding of the organization of the project.

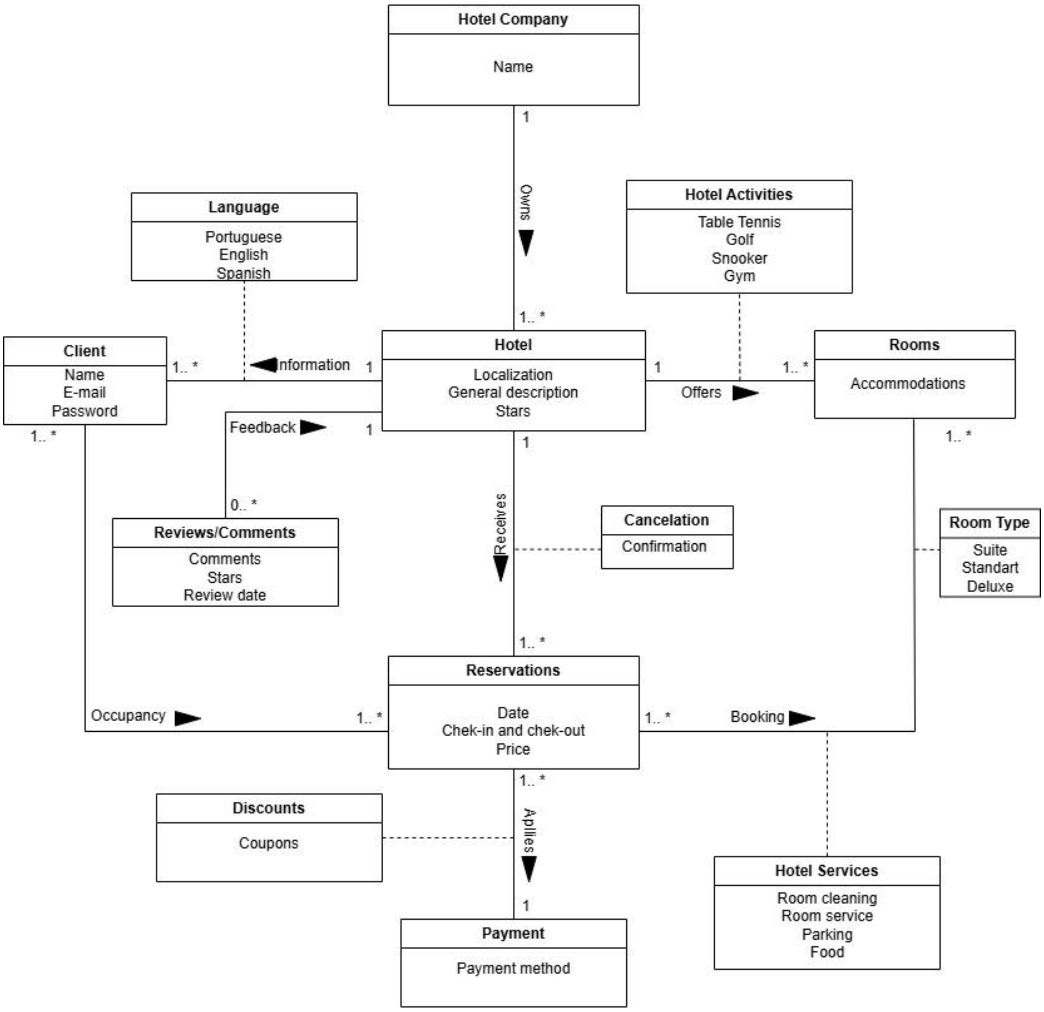
In addition to the redefined domain model, we have substantial advancements in the system's overall architecture and documentation. This report introduces both the physical and logical architectures, outlining the system's deployment structure and component-level organization. To ensure clarity and maintainability, the codebase is now structured according to well-defined layers and moules.

We also present a comprehensive suite of UML diagrams that document the internal structure and behavior of the system. These being a model class diagram, an exceptions class diagram, a views class diagram, a controllers class diagram, a sequence diagram and an user interface.

2. Changes

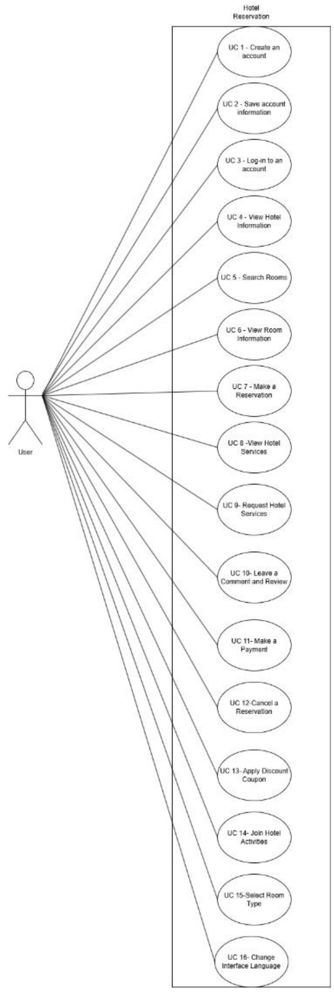
2.1. Domain Model

With the new Domain Model, it's easier to understand the intents of the Hotel Room Reservation System, as well as more complex, which is something that our first iteration was lacking.



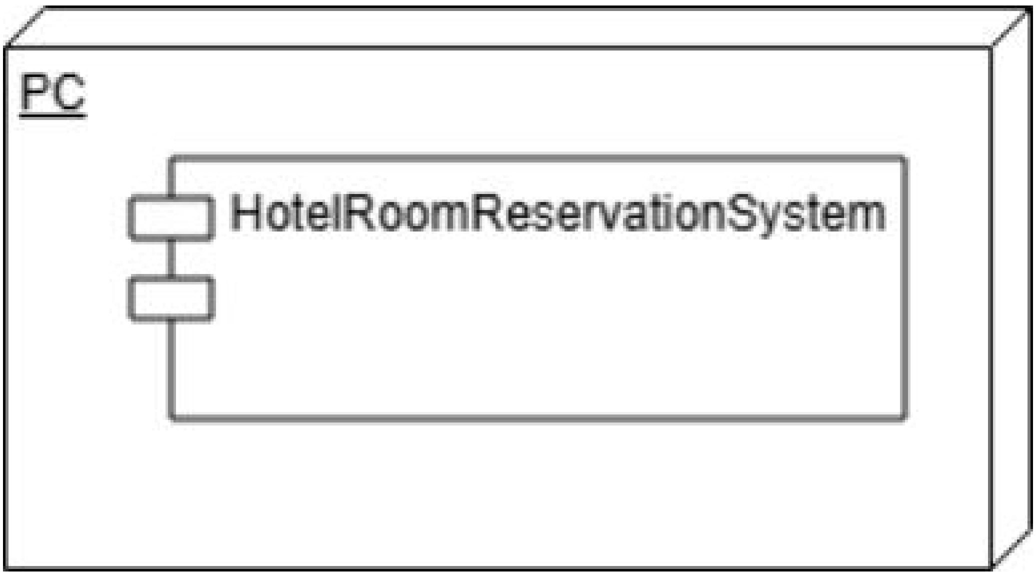
2.2. Functionalities

As a result of the domain model changes, as well as this iteration work, the functionalities of the system are now different. It is now clearer what each functionality does, as well as there is a better division of functionalities.

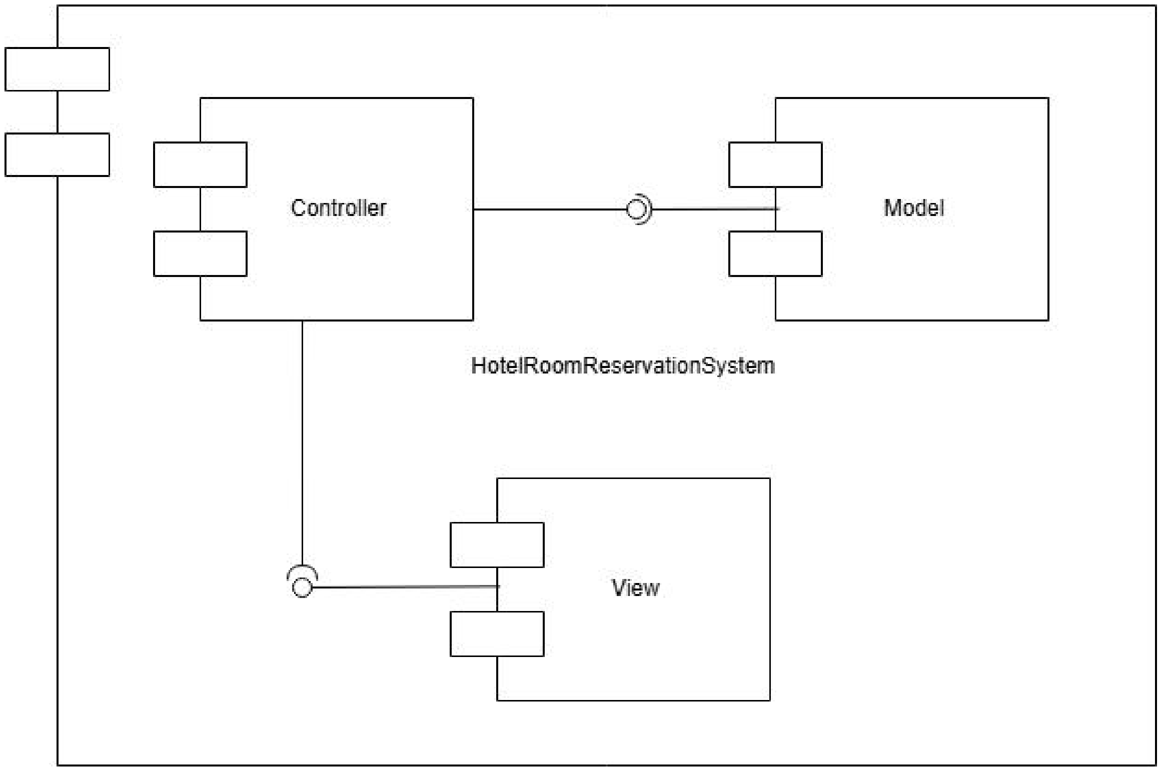


3. System Architecture

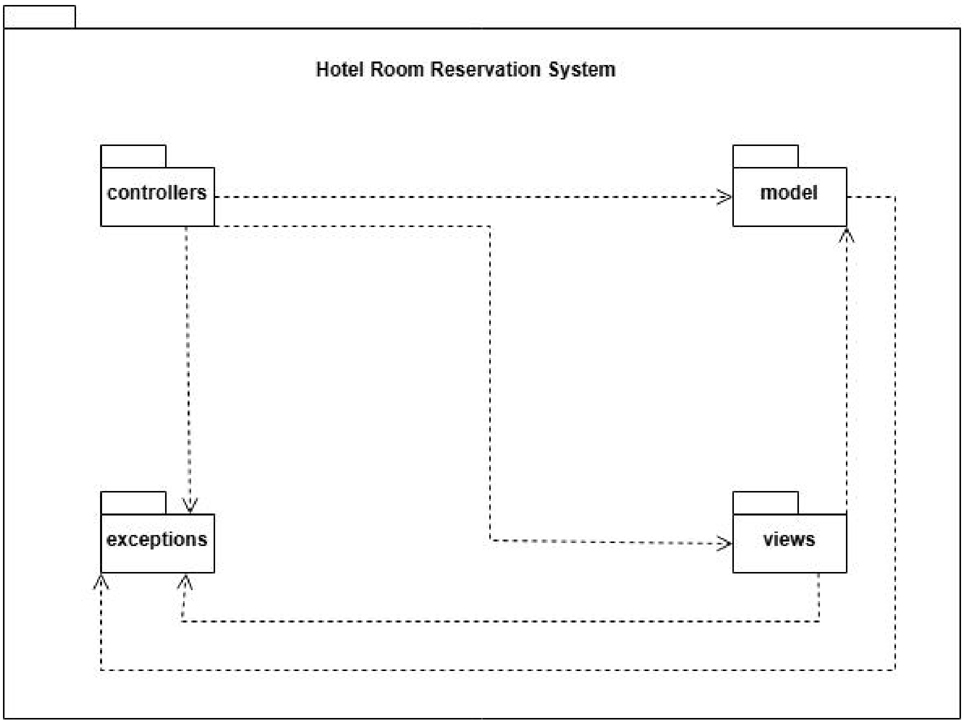
3.1. Physical Architecture



3.2. Logical Architecture



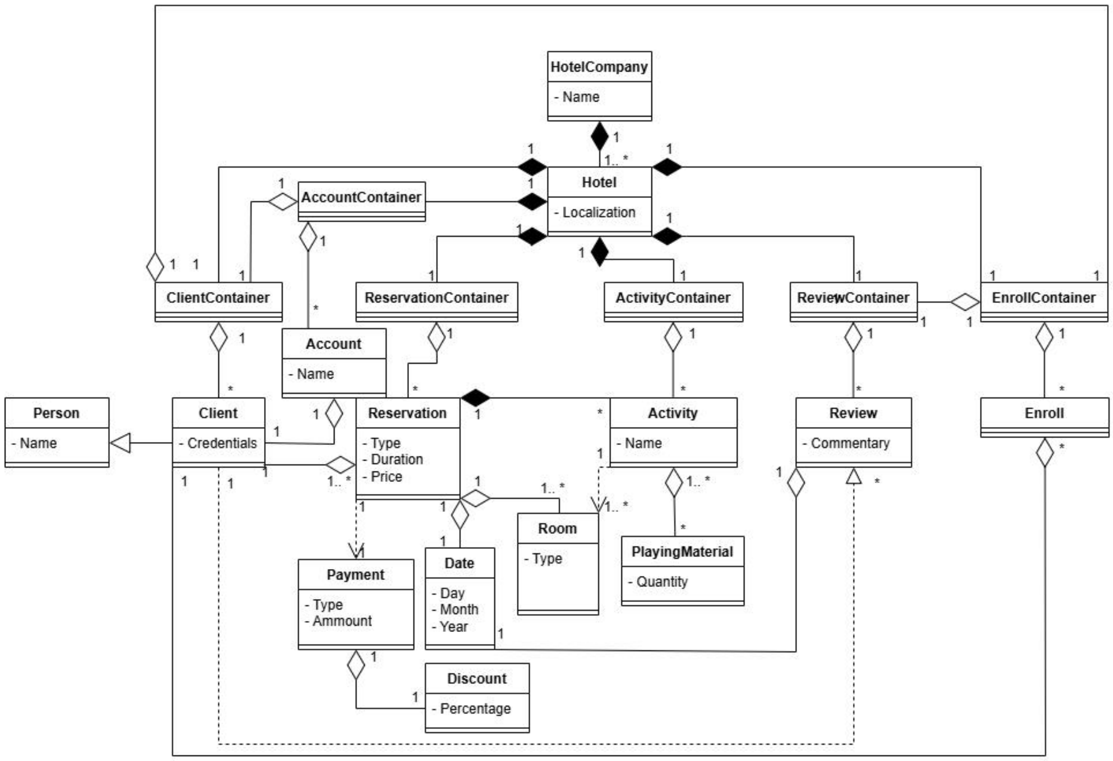
3.3. Code Organization



4. Class Diagrams

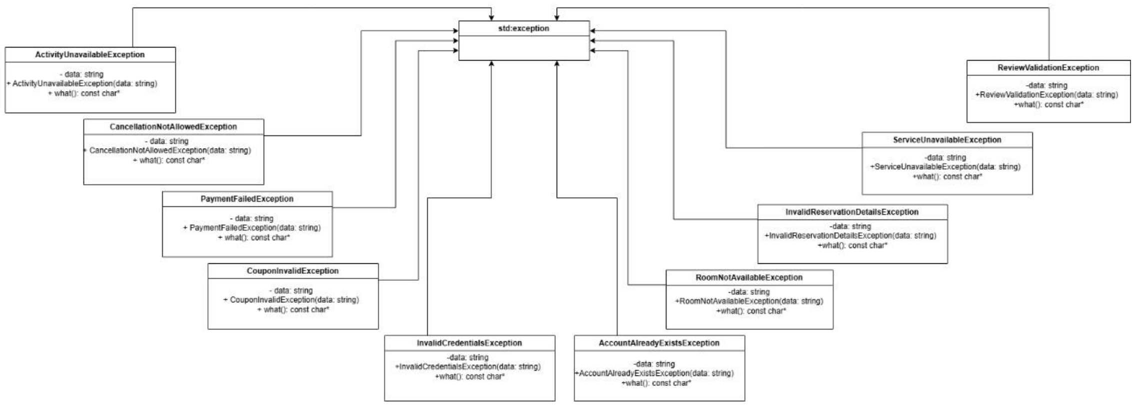
4.1. Model Class Diagram

The Model Class Diagram allows detailing of the core entities and their relationship.



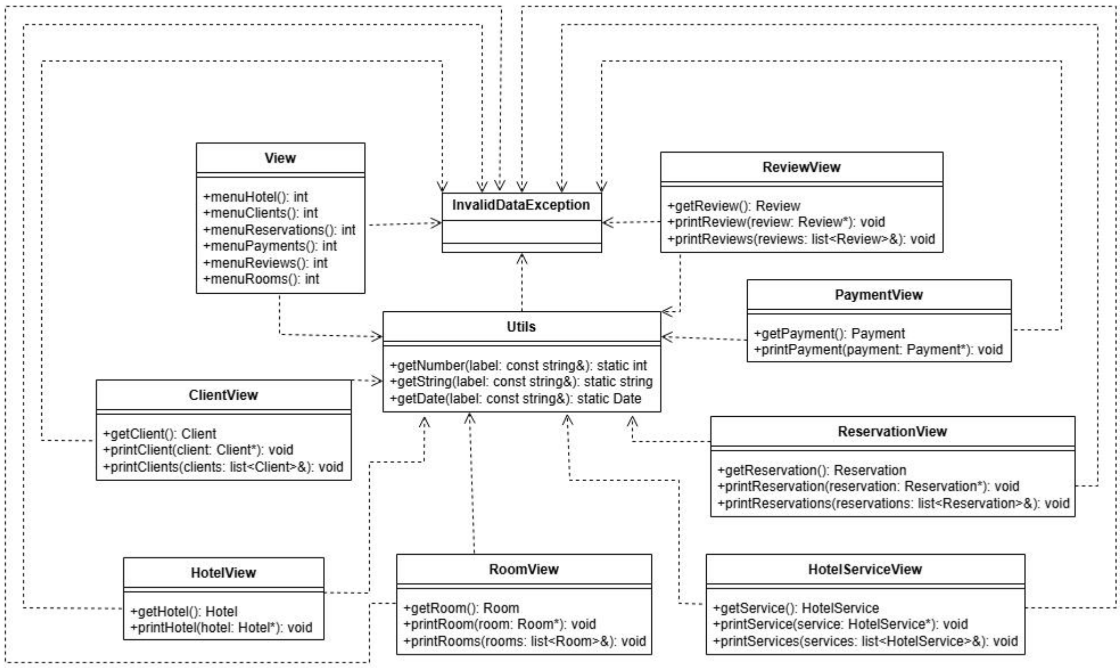
4.2. Exceptions Class Diagram

The Exceptions Class Diagram highlights custom error handling mechanisms.



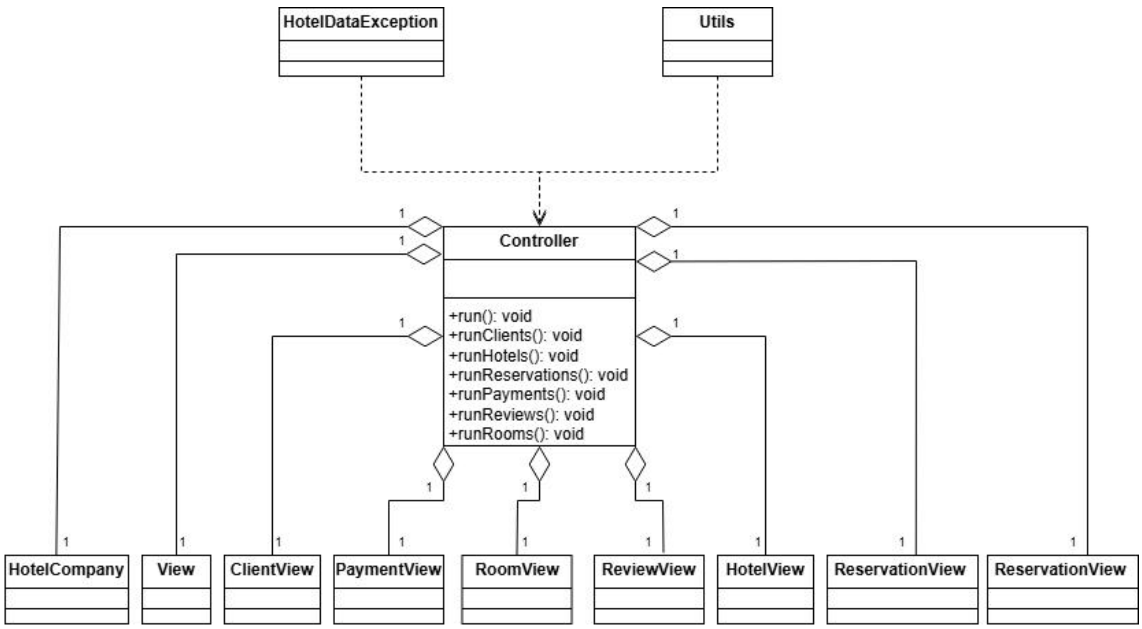
4.3. Views Class Diagram

The Views Class Diagram shows the user interface components.



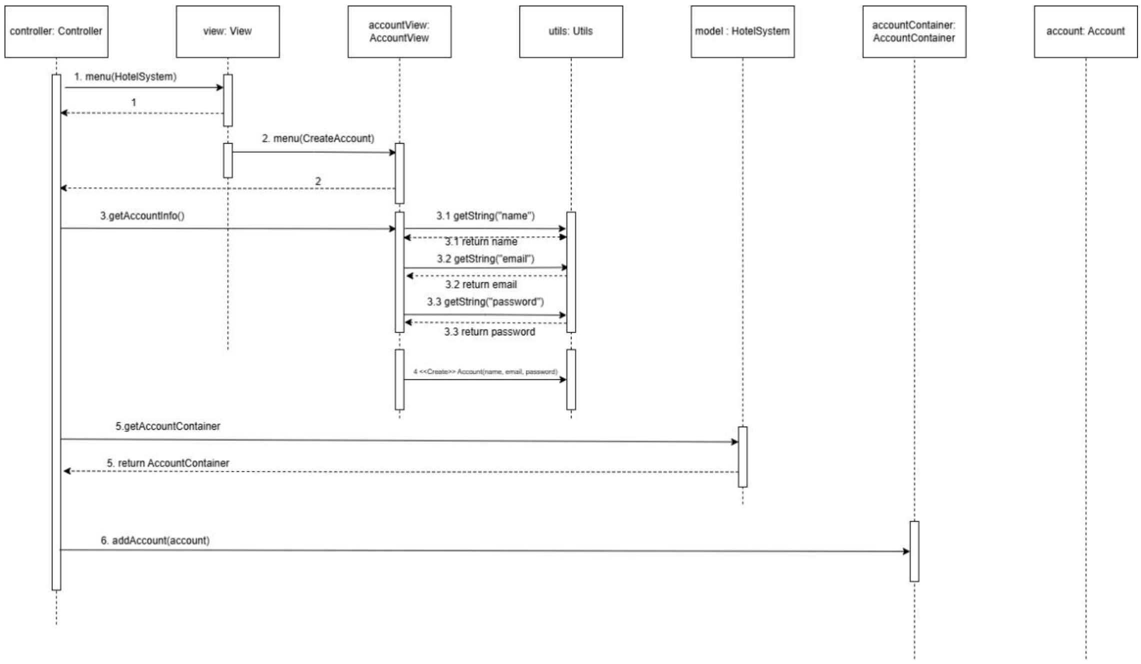
4.4. Controllers Class Diagram

The Controllers Class Diagram maps out the control flow logic.

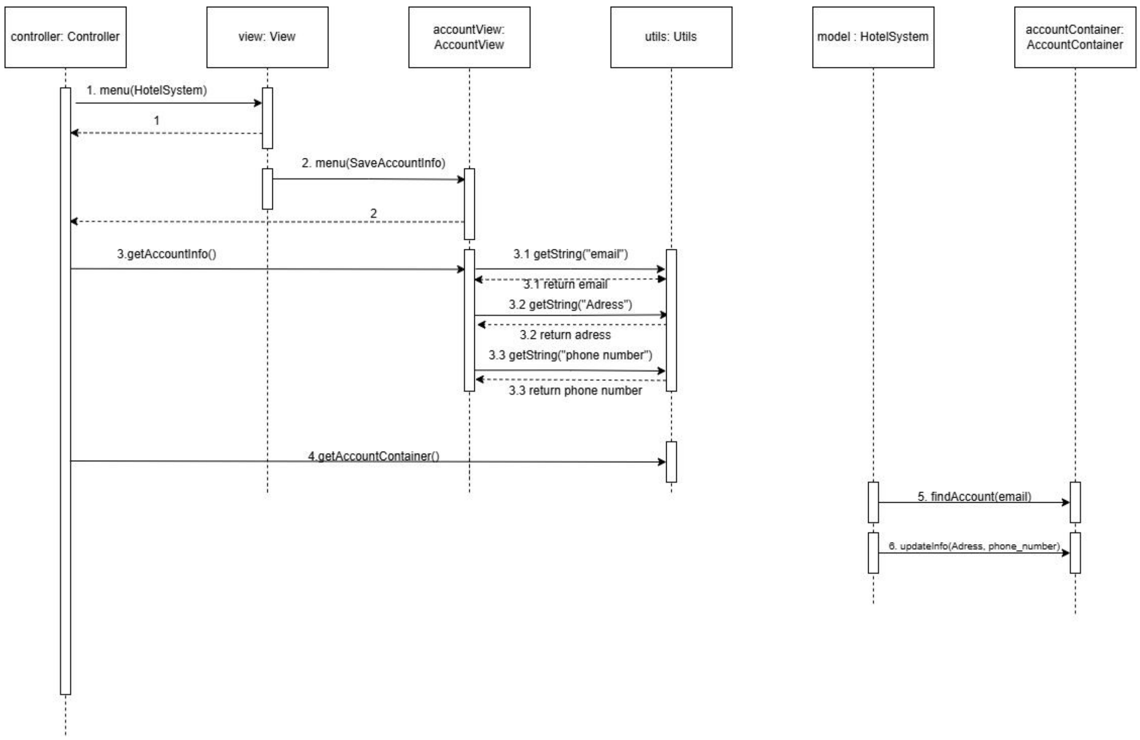


4. SSD’s

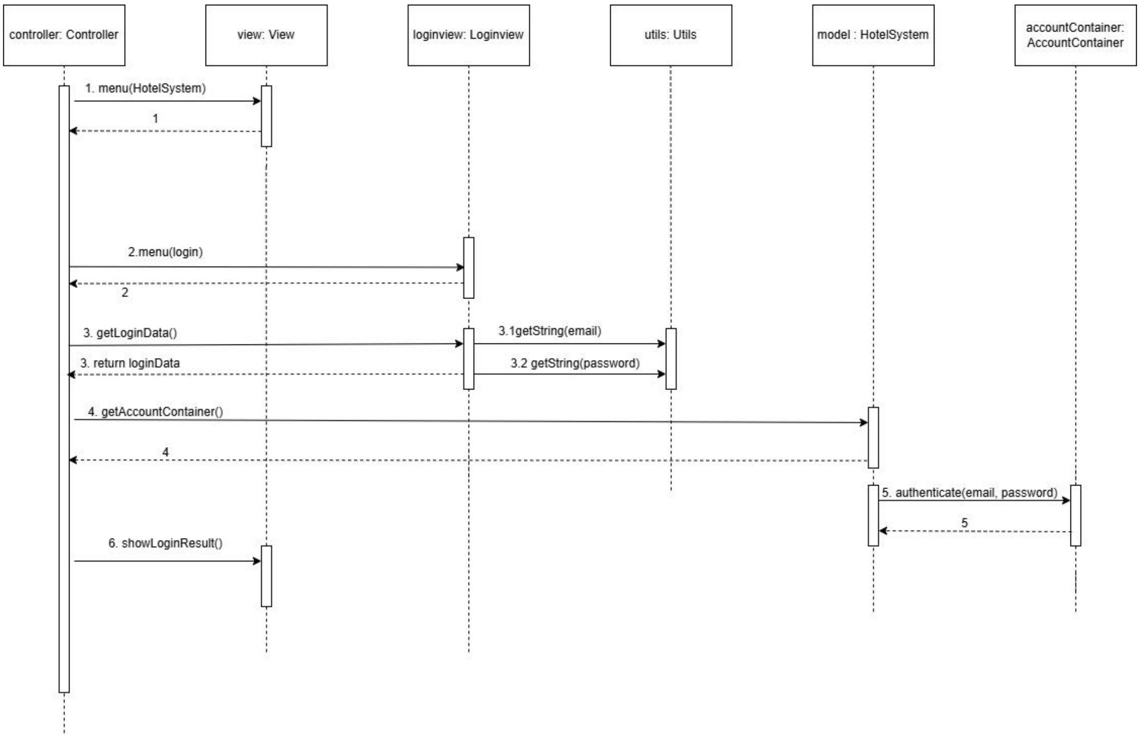
UC1 – Create an account:



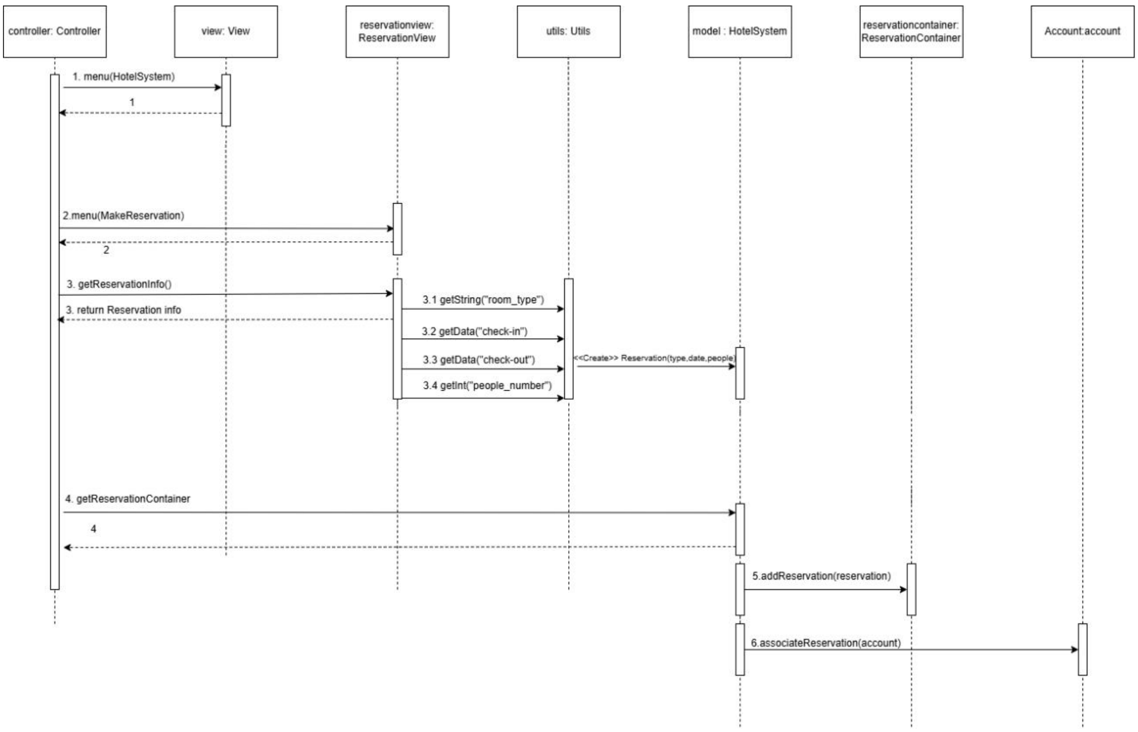
UC2 – Save account information:



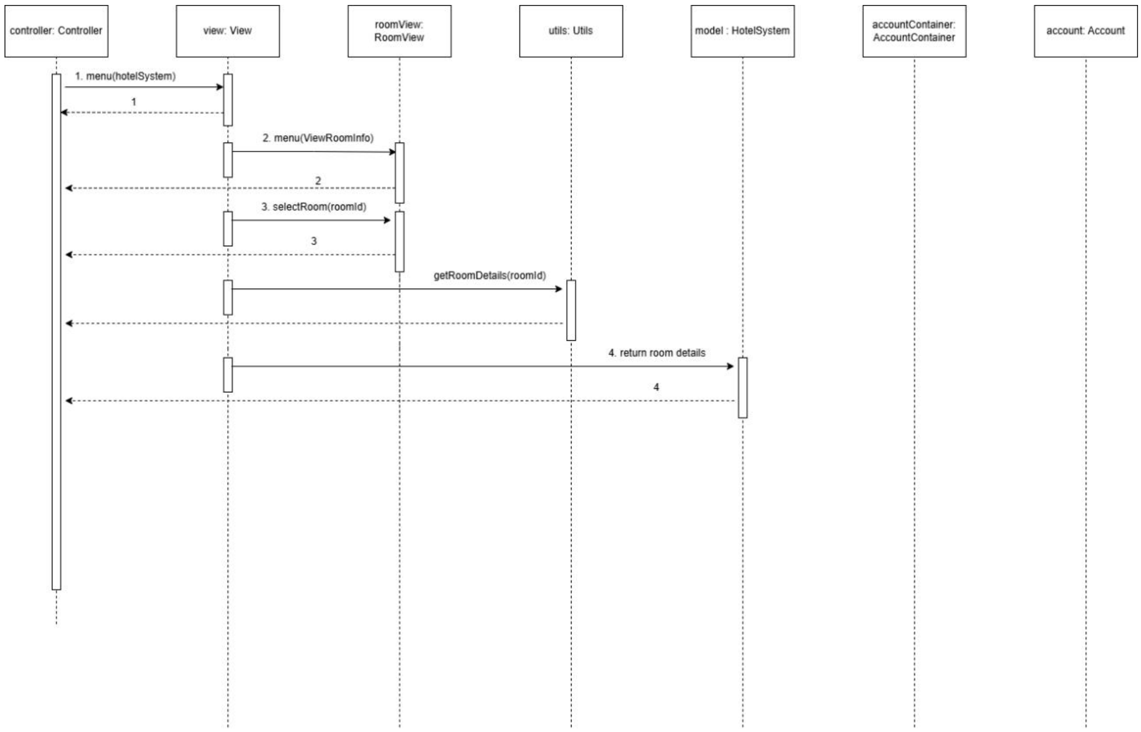
UC3 – Log-in to an account:



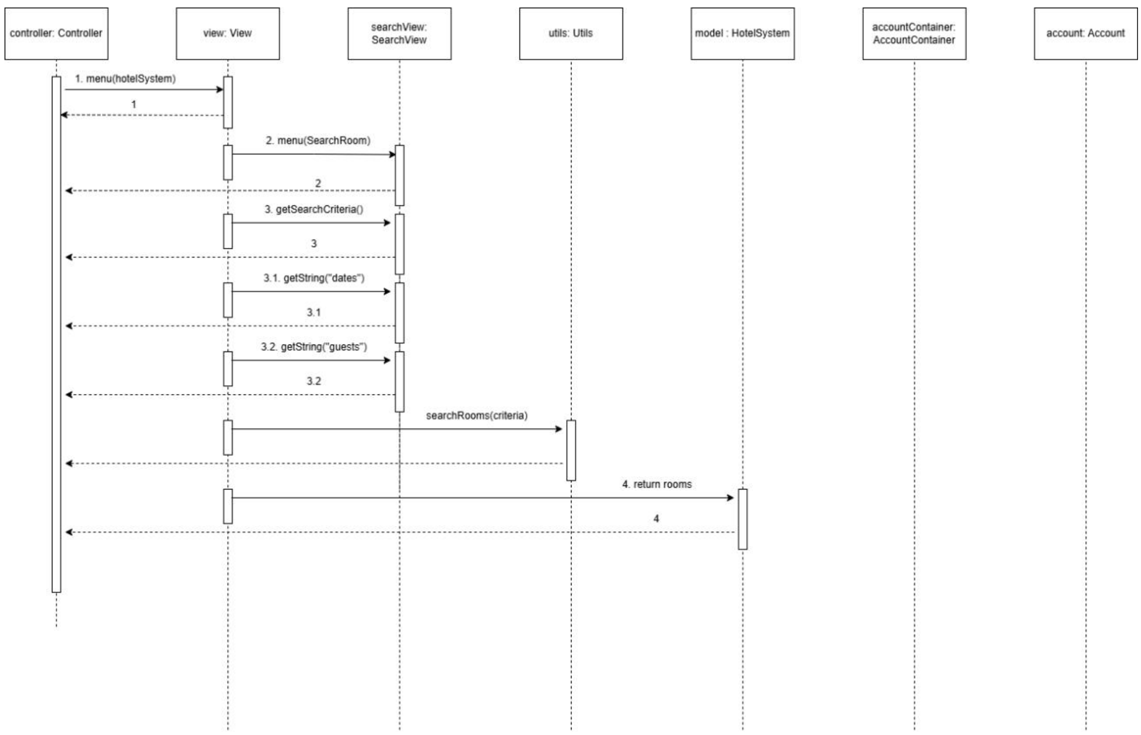
UC4 – View Hotel information:



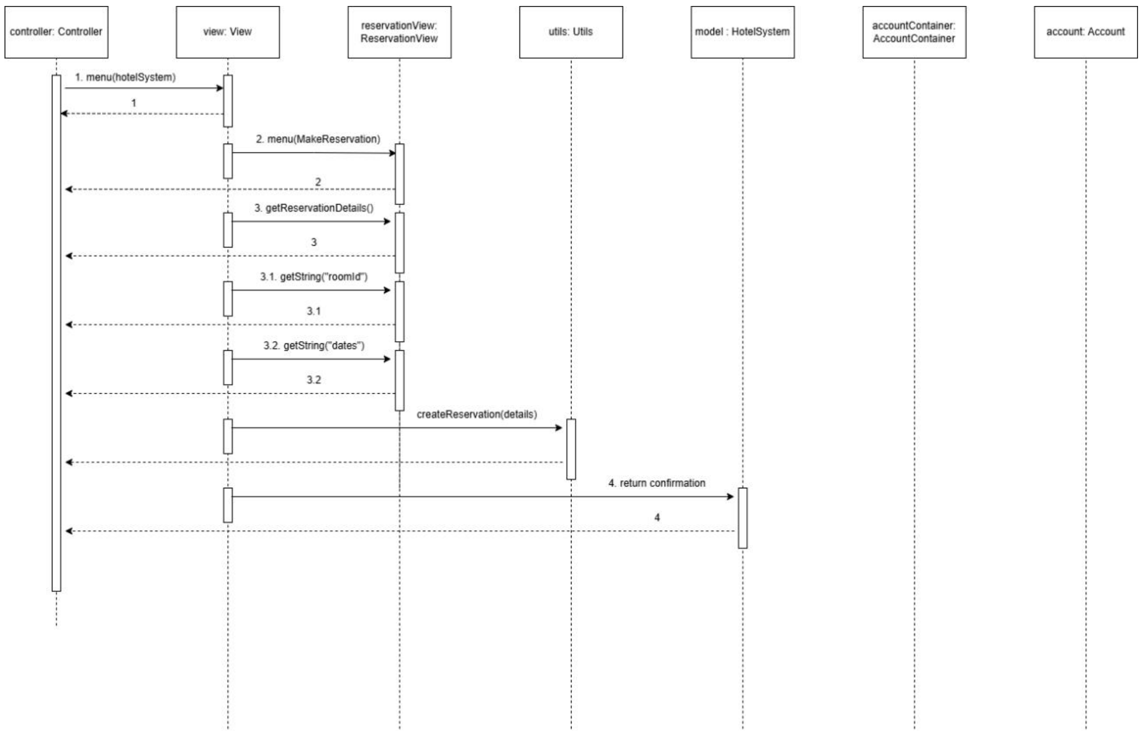
UC5 – Search rooms:



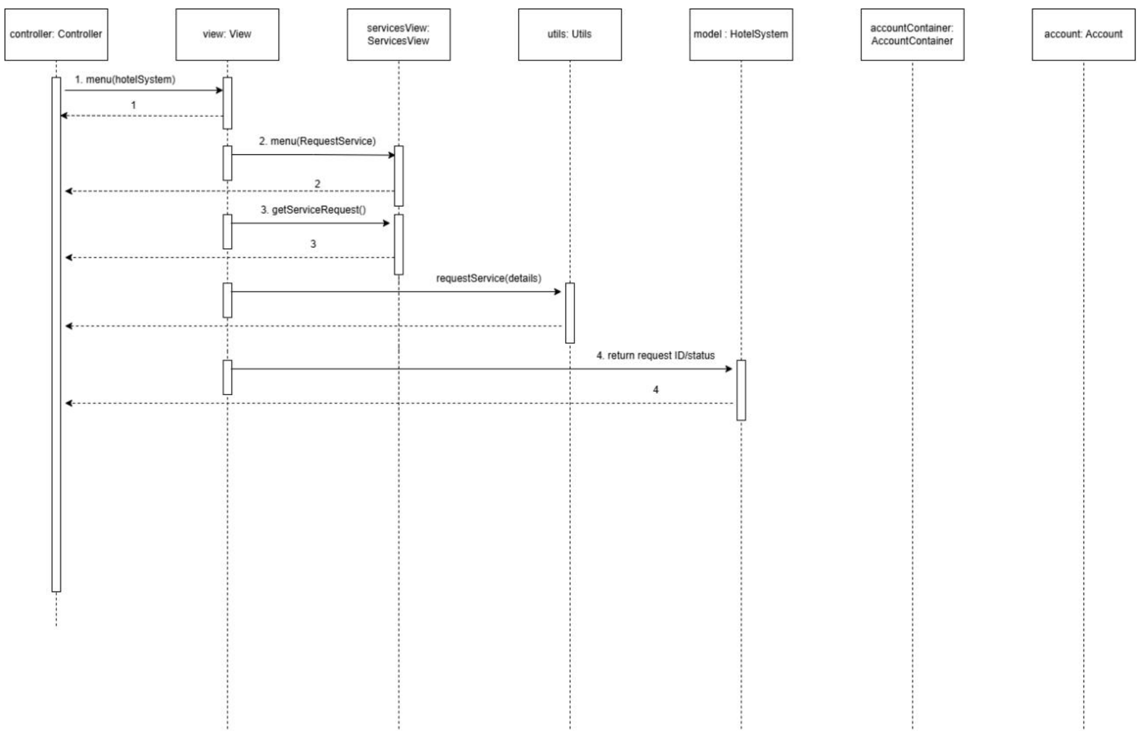
UC6 – View room information:



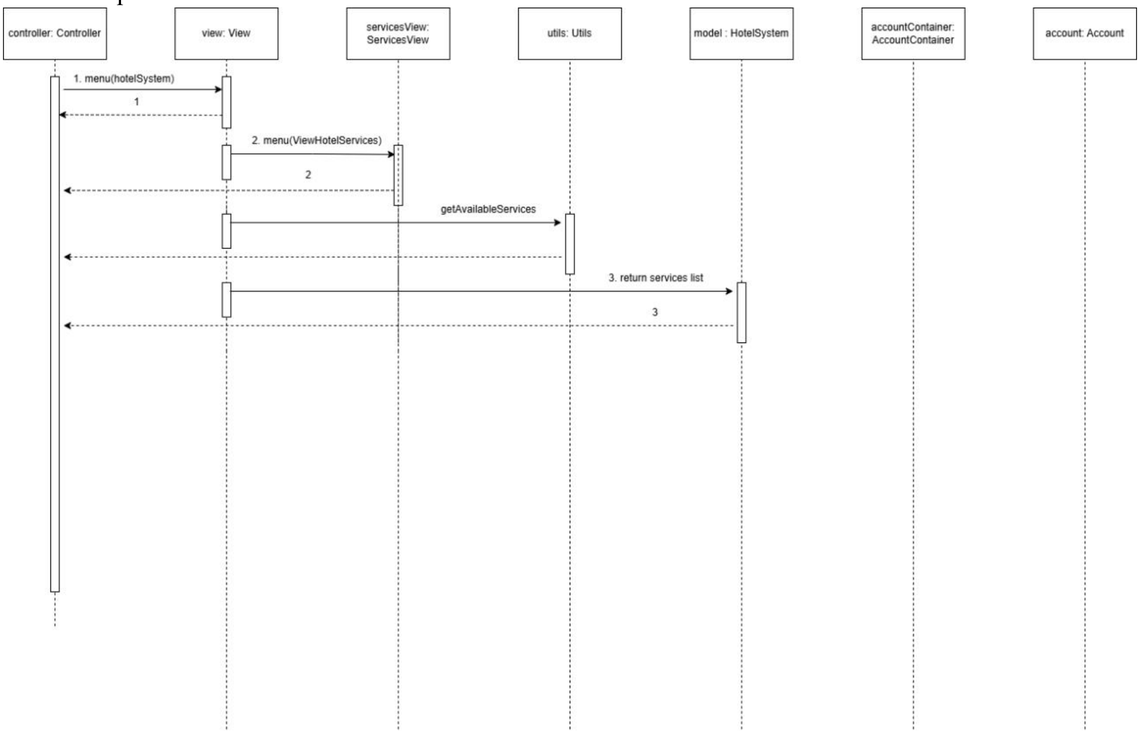
UC7 – Make a reservation:



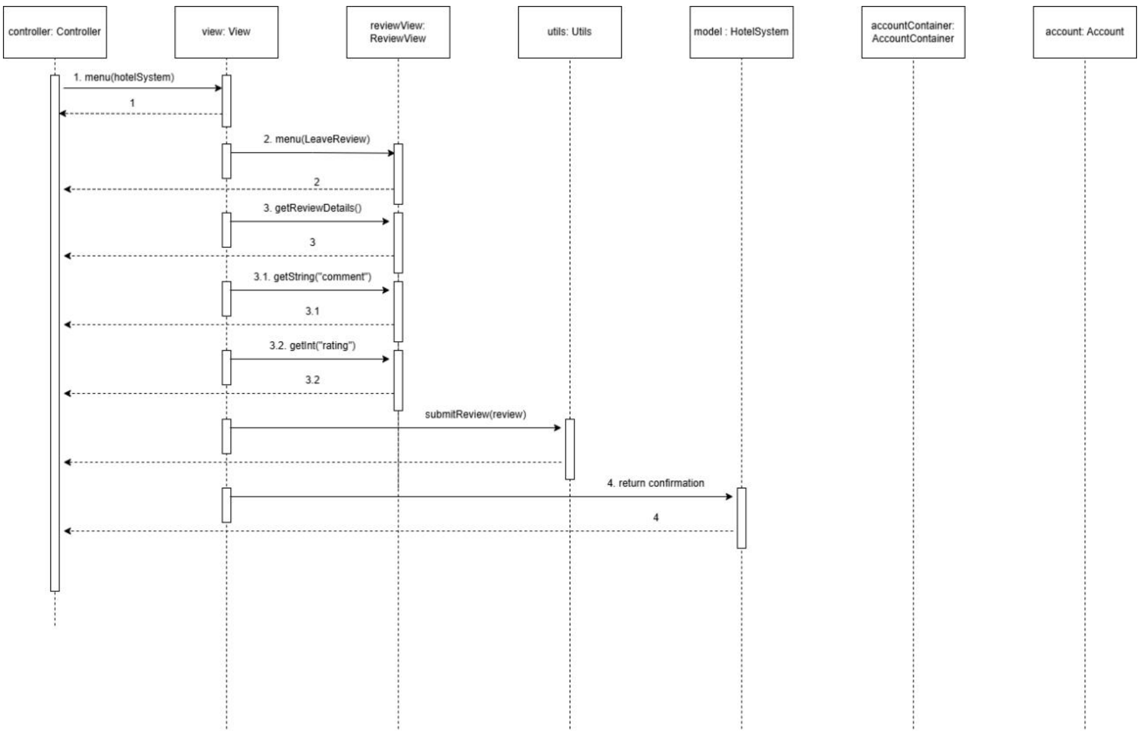
UC8 – View Hotel services:



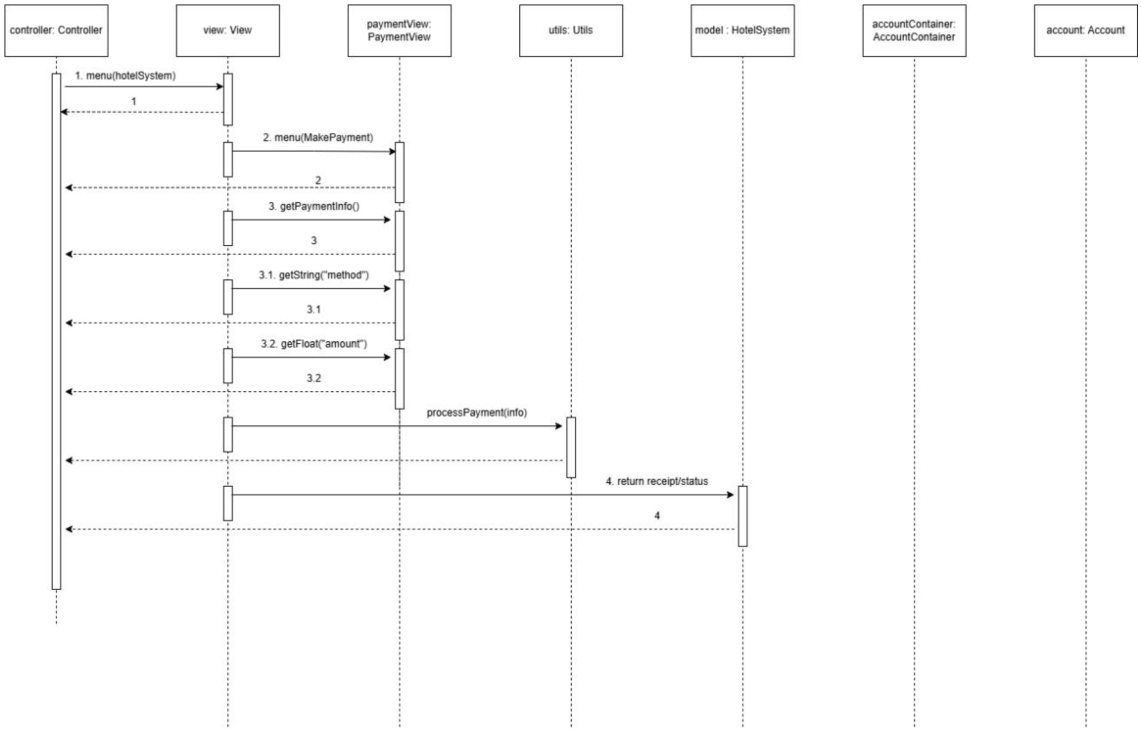
UC9 – Request Hotel services:



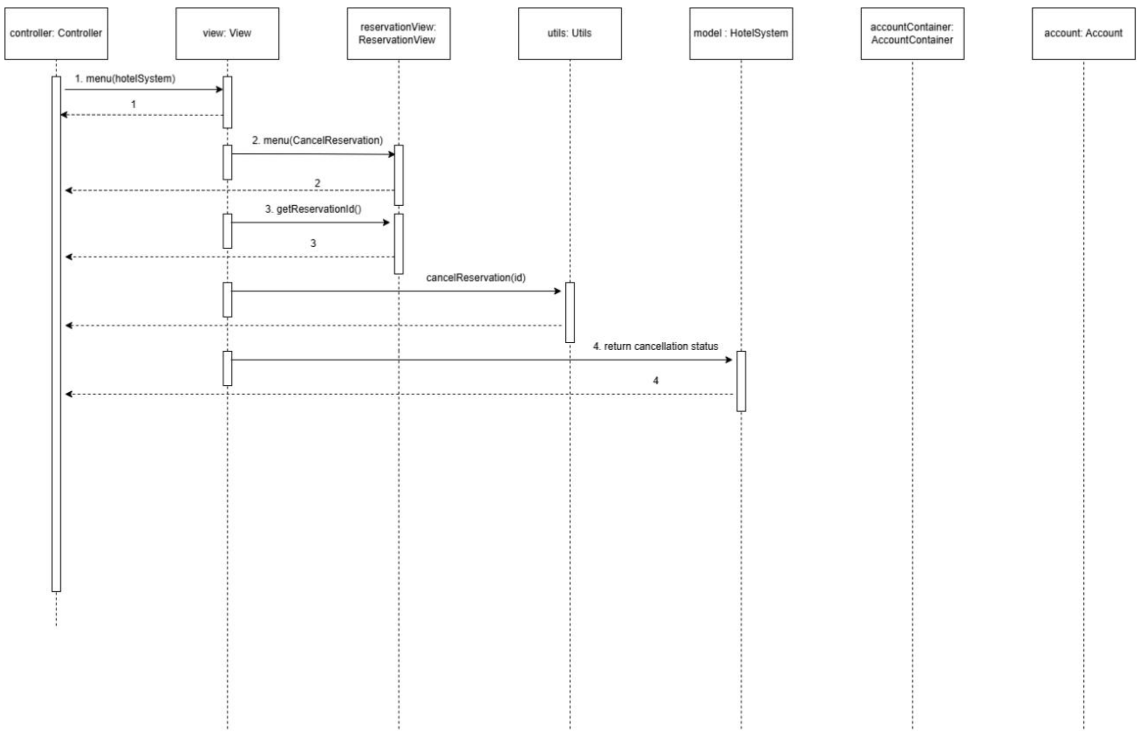
UC10 – Leave a comment and review:



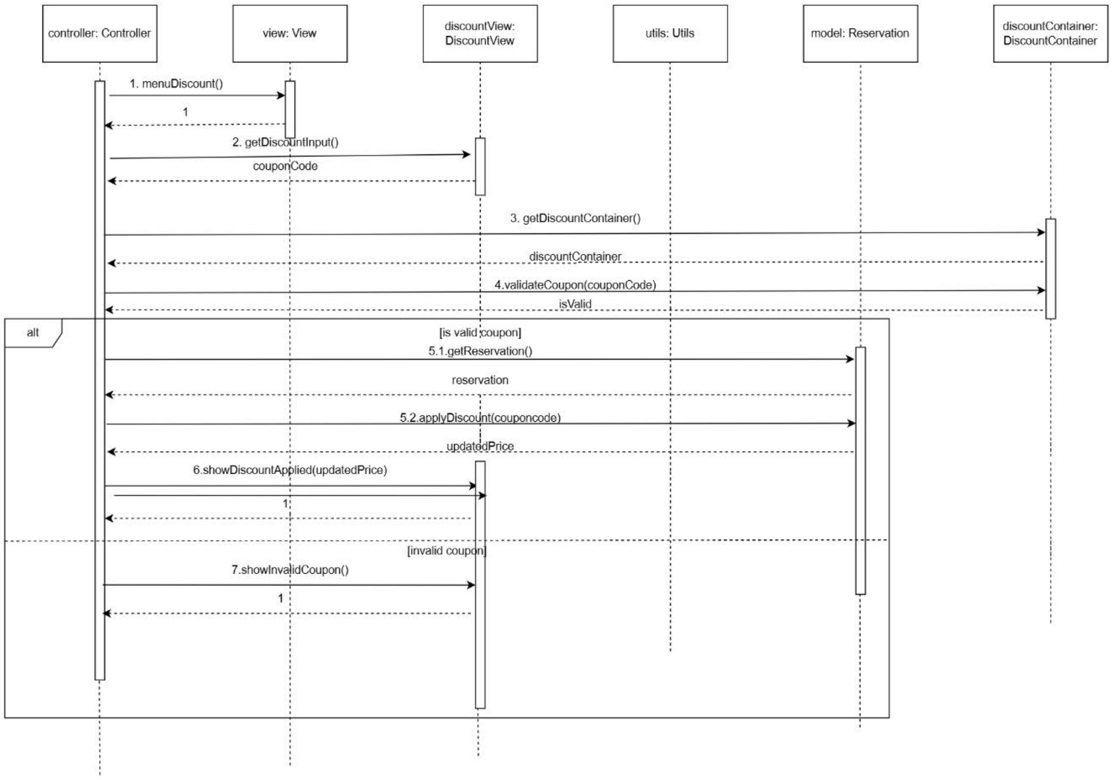
UC11 – Make a payment:



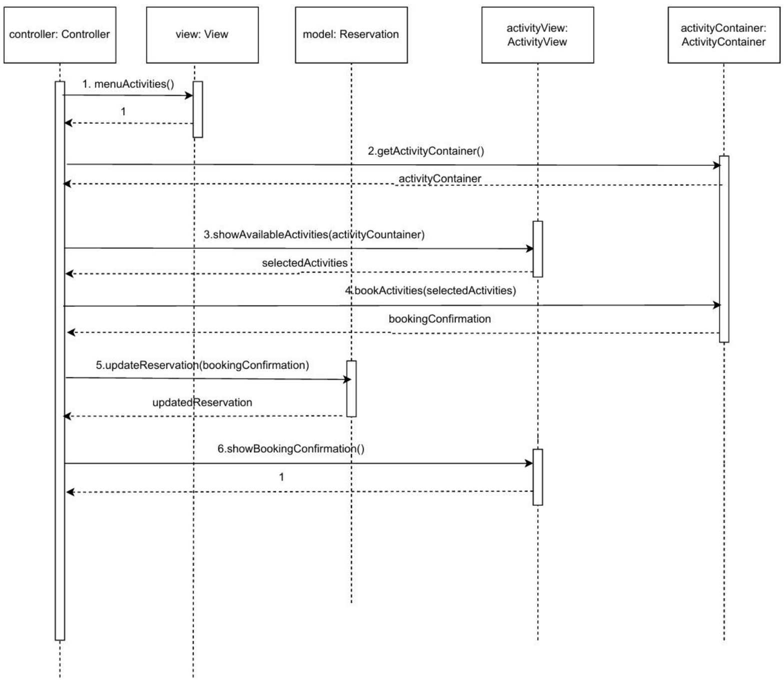
UC12 – Cancel a reservation:



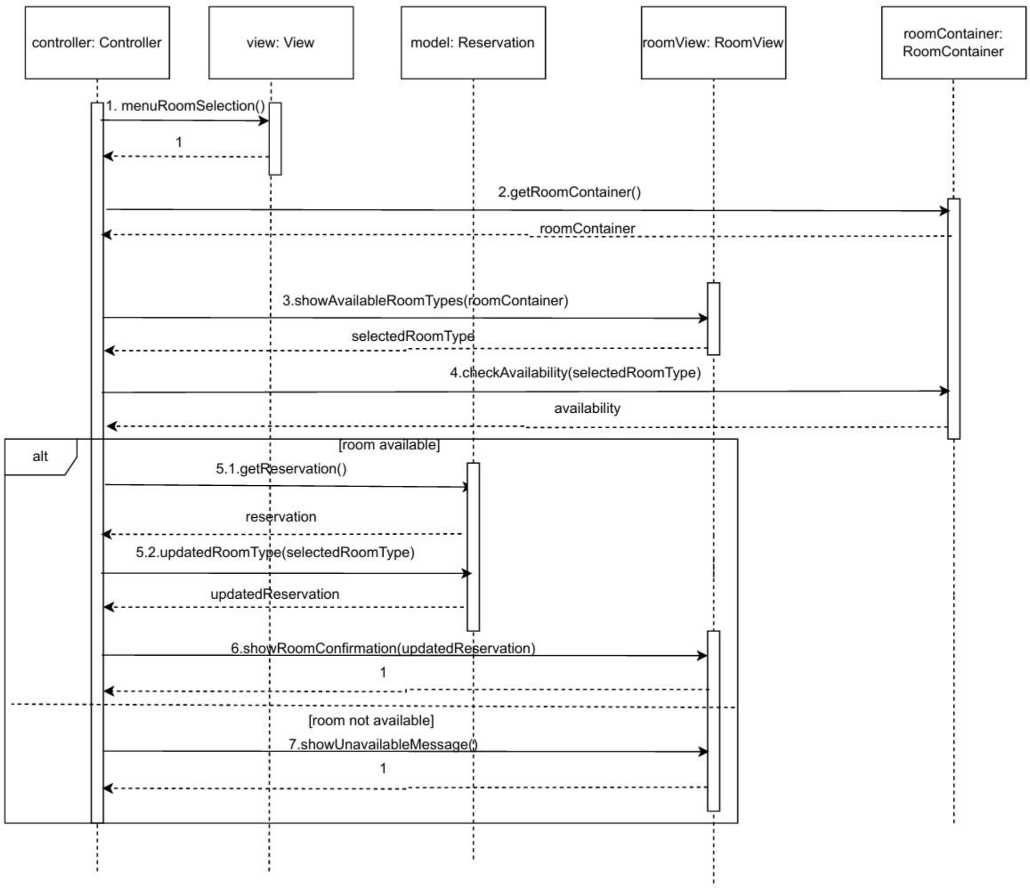
UC13 – Apply discount Coupon:



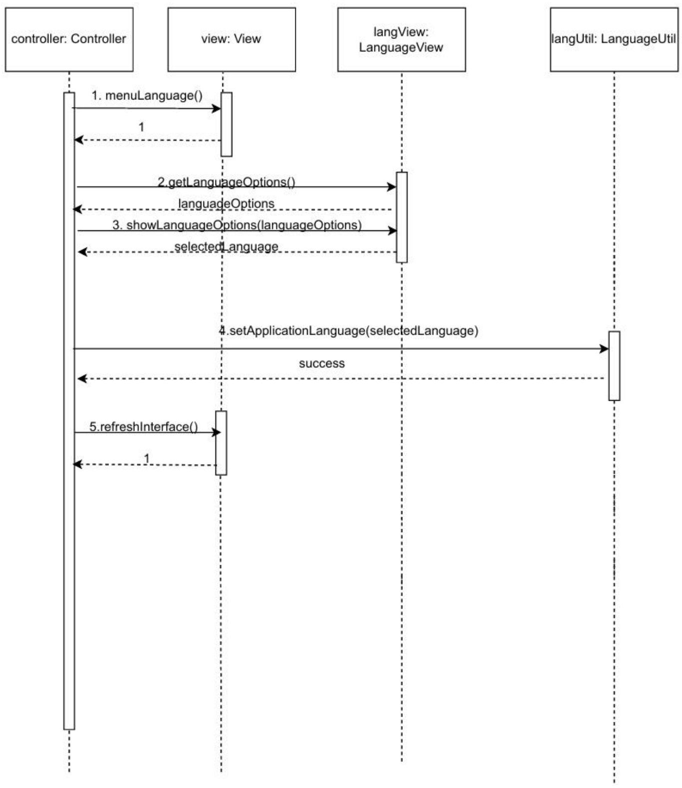
UC14 – Join Hotel activities:



UC15 – Select room type:



UC16 – Change interface language:



5. User Interface